Gujarati News Article Classification — Project Audit

An honest, evidence-based audit of the repository.

This document records what is **implemented and verified**, what is **planned or inferred**, and what **could not be verified** during the documentation pass. It exists so that a reader can immediately see the confidence level behind every claim in the handbook, and so that any discrepancies are visible rather than hidden.

Nothing in the repository was modified to produce this document.

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1. Audit method

Source inspected	How	Purpose
`config.py`, `src/*.py`, root scripts	read	ground-truth behaviour & contracts
`app/*.py`, `app/README.md`	read	GUI behaviour & honesty framing
`reports/**`	read	the project's own numeric evidence
`data/processed/*.meta.json`	read	artifact-level facts (row counts, OOV, fallbacks)
`data/raw/*.csv`	read-only	row counts, columns, class distribution
`tests/**`	counted & read	coverage of behaviour, side effects
File sizes under `models/`, `data/`	listed	disk footprint

Verification levels used below: *Verified* (read directly), *Inferred* (follows from verified facts), *Unverified* (could not confirm here).

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2. Implemented and verified

Capability	Status	Evidence
Dataset inspection (Phase 1)	Verified	`reports/dataset_inspection_report.txt`, `src/dataset_inspection.py`
Gujarati-safe preprocessing (Phase 2)	Verified	`reports/preprocessing_report.txt`, `src/preprocessing.py`, tests
Word2Vec CBOW + Skip-gram (Phase 3)	Verified	`reports/word2vec_training_report.txt`, `models/*.model`
Mean-pool document vectors (Phase 4)	Verified	`reports/document_vectors_report.txt`, `data/processed/*.npz` + `.meta.json`
12 classification experiments (Phase 5)	Verified	`reports/classification_validation_table.md`, `classification_experiments.csv`
Frozen test evaluation	Verified	`reports/classification_test_report.txt`
Single-inference facade	Verified	`src/prediction.py` (`NewsClassifier`)
Streamlit GUI (Phase 6)	Verified	`app/streamlit_app.py`, `app/content.py`, `app/README.md`
Error analysis (Phase 7)	Verified	`reports/error_analysis/*`
Improvement experiments + decision (Phase 7)	Verified	`reports/experiments/*`
Test suite	Verified	13 modules, **206** test functions counted
Centralised configuration	Verified	`config.py`

Key verified numbers

Quantity	Value	Source
Raw rows (train / valid / test)	79,416 / 10,251 / 10,041	inspection report
Cleaned rows	78,806 / 10,175 / 9,972	preprocessing report
Word2Vec vocabulary	210,459	training report
Vector dimension	100	training report
Selected experiment	exp008 (skip-gram docvec + gradient_boosting)	validation table
Validation macro-F1 (selected)	0.8883	validation table
Test macro-F1 / accuracy	0.8910 / 0.8953	test report
Baseline reproduction difference	0.00e+00	baseline reproduction report
Phase 7 improvement to baseline	+0.0006 (< 0.005 threshold → retain)	recommendation

3. Planned, inferred, or out of scope

Item	Status	Note
Deep-learning transformer model	Out of scope (by design)	explicitly a non-goal
REST API / database / deployment config	Not implemented	project is a demo, not a service
Multi-label classification	Not implemented	single-label by design
Hyper-parameter search beyond the fixed grid	Not implemented	planned grid only
`LDC` / `LPC` dataset formats	Not present	loader recognises them; only `SHC` is used
Cross-machine bit-exact reproduction	Unverified	Word2Vec `workers=8` makes it *likely*, not guaranteed
Exact count of currently *passing* tests	Unverified here	depends on data/artifacts present in the environment

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4. Discrepancies and observations

These are honest findings — some are harmless and expected, some are worth an action.

4.1 CSV filenames vs. actual columns

- **Finding:** filenames encode `SHC_LDC_LPC`, but the CSVs contain only `text` and `labels` (the SHC shape).
- **Impact:** none functionally; `DATA_FORMAT = "SHC"` matches the actual data.
- **Action:** cosmetic only — consider renaming or documenting the naming scheme.

4.2 No `.venv` present in the repository

- **Finding:** the project's own notes reference `.venv\Scripts\`, but no virtual environment existed during this pass; a system Python 3.11 was used.
- **Impact:** none on correctness; affects how a newcomer sets up the environment.
- **Action:** create `.venv` and install `requirements.txt` (Handb., Ch. 10).

4.3 Orphaned cleaned-text dumps

- **Finding:** `data/processed/*.preprocessed.csv` total ≈652 MB and are **not read** by any downstream stage (`generate_document_vectors()` re-derives tokens from `data/raw/`).
- **Impact:** wasted disk; no correctness impact.
- **Action:** safe to delete and regenerate on demand.

4.4 Very large classifier artifacts

- **Finding:** some joblib experiment bundles are hundreds of MB (e.g. exp011 ~822 MB) because the fitted vectorizer/SVD is embedded.
- **Impact:** disk usage in `models/classifiers/` and `models/experiments/`.
- **Action:** the selected model is comparatively small; large experiments could be pruned.

4.5 Scratch test report

- **Finding:** `reports/test_report.txt` contains only a short smoke-test string (scratch), not real results.
- **Impact:** could confuse a reader looking for "the test report".
- **Action:** the real test evidence is `classification_test_report.txt`.

4.6 Two tests overwrite report files

- **Finding:** two tests intentionally exercise real report writers and thus rewrite `reports/dataset_inspection_report.txt` and `reports/preprocessing_report.txt`.
- **Impact:** report contents change after a full test run; source CSVs are unaffected.
- **Action:** run those tests last, or restore reports afterwards.

4.7 Character n-gram TF-IDF disabled

- **Finding:** the classification report notes a char n-gram TF-IDF variant was disabled (~459 s fit/transform; ~74.5M non-zeros; >1.2 GB CSR on a 7.2 GB machine).
- **Impact:** the word-level TF-IDF baseline was used instead — a documented, resource-driven decision.
- **Action:** revisit only on a larger-memory machine.

4.8 Gensim sidecar files required

- **Finding:** Word2Vec `.model` files require their `.wv.vectors.npy` (and `.syn1neg.npy`) sidecars; copying only the `.model` breaks `Word2Vec.load`.
- **Impact:** a common deployment pitfall.
- **Action:** ship the sidecars with the model.

4.9 Not a Git repository

- **Finding:** the workspace was not a Git repository during this pass, though a `.gitignore` is present.
- **Impact:** none; version history is simply not available to inspect.
- **Action:** initialise Git and commit only source/reports (large binaries are already ignored).

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5. Risk register

Risk	Likelihood	Impact	Mitigation
Accidental test-set leakage in future changes	Low	High	keep `select_best` /TF-IDF fit train(+valid) only; re-read Ch. 7
Missing model sidecars on deployment	Medium	High	ship `.npy` sidecars; GUI already errors helpfully
Disk exhaustion from large artifacts	Medium	Medium	prune unused experiment bundles
Cross-machine non-determinism	Low	Low	document; pin versions; treat as expected
Confusion between fuzzy categories persists	High	Low	intrinsic to task; documented, not a bug

6. Audit verdict

- The **core pipeline is fully implemented and internally consistent**, and its reported numbers are corroborated across multiple independent artifacts (validation table, test report, baseline reproduction, Phase 7 comparison).
- **Evaluation discipline is strong**: validation-only selection, a single frozen test pass, and an explicit retain/replace threshold.
- The **remaining issues are operational/cosmetic**, not correctness problems: orphaned CSV dumps, oversized artifacts, a scratch `test_report.txt`, and the absent virtual environment.
- Nothing was found that contradicts the project's headline claims (test macro-F1 0.8910, accuracy 0.8953).

End of audit.